

Replication: Carbon Taxes and CO2 Emissions

Synthetic control, illustrated with Andersson (American Economic Journal: Economic Policy, 2019)

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1 The paper in one page

The question is whether a carbon tax reduces emissions, answered with one country and thirty years of history. Sweden extended its value-added tax to transport fuel in 1990 and introduced a carbon tax on it in 1991, on top of an existing energy tax. Andersson (2019) asks what happened to per-capita CO2 emissions from transport, the sector the tax covers. The answer has to come from a comparison, and the natural comparison, the average of other OECD countries, does not trend like Sweden before 1990.

The design is a synthetic control with a donor pool chosen by hand and weights chosen by the data. From 25 OECD countries the paper keeps 14 donors: it drops the countries that introduced carbon taxes on transport (Finland, Norway, the Netherlands) or made large fuel-tax changes (Germany, Italy, the United Kingdom), two countries with fuel tourism (Austria, Luxembourg), and two whose development is unlike Sweden's (Turkey, Ireland). Synthetic Sweden is the convex combination of the 14 donors that best reproduces Sweden's 1980 to 1989 averages of GDP per capita, motor vehicles, gasoline consumption, and urban population, and its emissions in 1970, 1980, and 1989. The weights on the predictors are chosen jointly with the country weights to minimize the pre-treatment mean squared prediction error of emissions over 1960 to 1989.

The headline is a reduction of almost 11 percent in an average post-treatment year. Emissions in Sweden are 0.29 metric tons per capita below synthetic Sweden on average over 1990 to 2005, and 0.35 tons, or 12.5 percent, below it in 2005. The paper then splits the effect between the VAT and the carbon tax with a time-series regression of gasoline consumption on the tax-exclusive price and the carbon tax, finds that consumers respond about three times more strongly to the carbon tax than to an equivalent price change, and attributes 6.3 percent of the 10.9 percent to the carbon tax alone.

This notebook reproduces every table and figure from the replication data, adds the checks of Ferman and Pinto (2021), and reads each object as the paper does. Section 2 describes the data. Section 3 reproduces the descriptive figures and the pass-through regression. Section 4 builds synthetic Sweden (Tables 1 and 2, Figures 4 and 5). Section 5 runs the placebo tests (Figures 6 to 9). Section 6 checks the recession (Figures 11 and 12). Section 7 reproduces Table 3 and Figures 13 and 14. Section 8 adds the imperfect-fit diagnostics. All estimation uses `synth_control()` and its companions from `causalmetrics`; the regressions use `lm()`, `ivreg()`, and Newey-West standard errors from `sandwich`.

2 Data

The panel is 15 countries over 1960 to 2005 with the outcome and five predictors. The file `carbontax_data.dta` comes from the replication package as redistributed in the RTutor problem set of the paper. The outcome is CO2 emissions from transport in metric tons per capita (World Bank). The predictors are GDP per capita (PPP, 2005 dollars), motor vehicles per 1,000 people, gasoline consumption per capita (kilograms of oil equivalent), urban population (percent), and population density. Poland's GDP is missing before 1970, which matters only for the 1970 in-time placebo, where the paper's code drops Poland.

```
carbon <- as.data.frame(read_dta(file.path(data_path, "carbontax_data.dta")))
carbon$treated <- as.integer(carbon$country == "Sweden" & carbon$year >= 1990)
donors <- sort(setdiff(unique(carbon$country), "Sweden"))
xs <- c("GDP_per_capita", "vehicles_capita", "gas_cons_capita", "urban_pop")
x_labels <- c(GDP_per_capita = "GDP per capita", vehicles_capita = "Motor vehicles (per 1,000 people)",
              gas_cons_capita = "Gasoline consumption per capita", urban_pop = "Urban population",
              `CO2_transport_capita(1989)` = "CO2 from transport per capita 1989",
              `CO2_transport_capita(1980)` = "CO2 from transport per capita 1980",
              `CO2_transport_capita(1970)` = "CO2 from transport per capita 1970")
c(countries = length(unique(carbon$country)), years = length(unique(carbon$year)),
  pre_years = sum(unique(carbon$year) < 1990), post_years = sum(unique(carbon$year) >= 1990))
#>   countries      years pre_years post_years
#>      15         46      30         16
donors
#> [1] "Australia"      "Belgium"      "Canada"      "Denmark"
#> [5] "France"          "Greece"      "Iceland"     "Japan"
#> [9] "New Zealand"     "Poland"      "Portugal"    "Spain"
#> [13] "Switzerland"     "United States"
```

Three further files serve the descriptive figures and the second analysis. `descr_Sweden.Rds` holds the Swedish price and tax components and the fuel consumption series (Figures 1 and 2 of the paper). `regression_data.Rds` holds the annual Swedish series for 1970 to 2015 used in the pass-through regression and in Table 3: nominal and real prices, the energy and carbon taxes with and without VAT, the crude oil price, log gasoline consumption per capita, GDP per capita, urbanization, and unemployment. `disentangling_data.dta` holds the simulated emission paths under the three tax scenarios of Figure 13 and the synthetic control gap for Figure 14.

```
descr <- readRDS(file.path(data_path, "descr_Sweden.Rds"))
reg <- readRDS(file.path(data_path, "regression_data.Rds"))
disent <- as.data.frame(read_dta(file.path(data_path, "disentangling_data.dta")))
c(descr = nrow(descr), regression = nrow(reg), regression_complete = sum(!is.na(reg$log_gas_cons)), disentangling = nrow(disent))
#>           descr      regression regression_complete      disentangling
#>           46         46         42         46
```

3 Descriptive evidence and tax incidence

Figure 1 of the paper shows that the tax reform changed the price of fuel, which is the mechanism the whole design relies on. The energy tax existed before 1990; the VAT and the carbon tax were added in 1990 and 1991, and the carbon tax rose sharply after 2000 while the energy tax was reduced. The figure is there so the reader sees a large, dated change in the treatment variable.

```
descr %>% select(year, en_tax, CO2_tax, total_tax) %>%
  pivot_longer(-year, names_to = "component", values_to = "value") %>%
  mutate(component = recode(component, en_tax = "Energy tax", CO2_tax = "Carbon tax", total_tax = "Total tax")) %>%
  ggplot(aes(x = year, y = value, colour = component, linetype = component)) +
  geom_line() + tax_line +
  labs(x = NULL, y = "Real tax, SEK per liter", colour = NULL, linetype = NULL) + rep_theme()
```

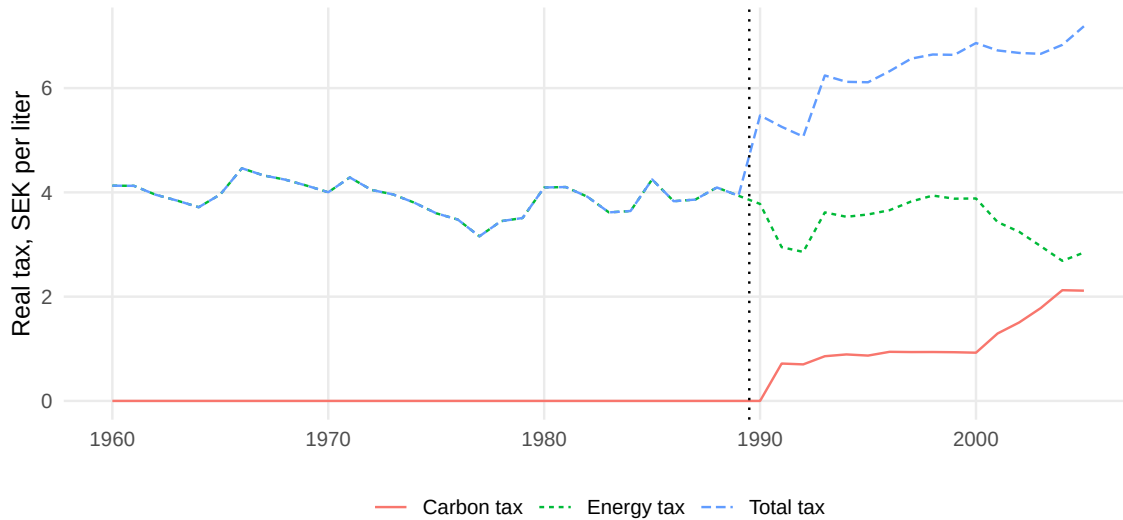


Figure 1: Real energy tax, carbon tax, and total tax on gasoline in Sweden, SEK per liter (paper's Figure 1, panel A).

Figure 3 of the paper is the reason for the method: Sweden and the donor average do not share a trend. Before 1980 the two series grow at a similar pace; between 1980 and 1989 Sweden's emissions grow twice as fast. A difference-in-differences estimate would attribute that trend difference to the tax. The figure is placed before any estimation so that the reader accepts the need for a weighted comparison.

```
donor_avg <- carbon %>% filter(country != "Sweden") %>% group_by(year) %>%
  summarise(y = mean(CO2_transport_capita), .groups = "drop") %>% mutate(series = "OECD sample")
sweden <- carbon %>% filter(country == "Sweden") %>% transmute(year, y = CO2_transport_capita, series = "Sweden")
plot_paths <- function(d, ylab = "Metric tons per capita (CO2 from transport)") {
  ggplot(d, aes(x = year, y = y, colour = series, linetype = series)) +
    geom_line() + tax_line +
    scale_colour_manual(values = c("black", "grey45", "#B22222")[seq_along(unique(d$series))]) +
    scale_y_continuous(limits = c(0, 3)) +
    labs(x = NULL, y = ylab, colour = NULL, linetype = NULL) + rep_theme()
}
plot_paths(bind_rows(sweden, donor_avg))
```

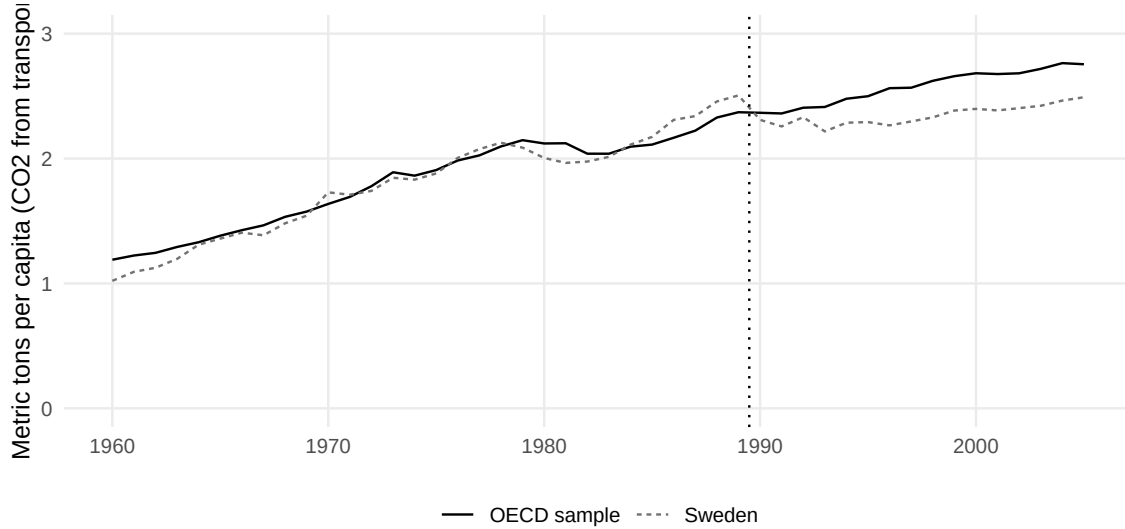


Figure 2: Per-capita CO2 emissions from transport, Sweden versus the unweighted average of the 14 donor countries (paper's Figure 3).

The pass-through regression (paper's equation 2) establishes that tax changes reach consumers, so that tax elasticities can be read as demand elasticities. The nominal retail price is regressed in first differences on the crude oil price and the sum of the energy and carbon taxes, 1970 to 2015. The paper reports a coefficient of 1.15 on taxes with a 95 percent interval of 0.90 to 1.39.

```
pass <- reg %>% arrange(year) %>%
  mutate(delta_p = p_nom - lag(p_nom), delta_oil = oil_p - lag(oil_p), delta_tax = en_CO2_tax - lag(en_CO2_tax))
pass_fit <- lm(delta_p ~ delta_oil + delta_tax, data = pass)
cbind(estimate = coef(pass_fit), confint(pass_fit))[c("delta_oil", "delta_tax"), ]
#>           estimate      2.5 %      97.5 %
#> delta_oil 0.006217283 0.004870472 0.007564094
#> delta_tax 1.147292386 0.638541836 1.656042936
```

The point estimate on taxes is the paper's 1.15. The interval above uses ordinary least-squares standard errors and is wider than the paper's; the paper does not state which standard errors it used for this regression, and the replication package's code for it is not part of the RTutor data.

4 Synthetic Sweden

The synthetic control is built exactly as in the paper: four covariates averaged over 1980 to 1989, three emission lags, and predictor weights chosen to minimize the pre-treatment MSPE over 1960 to 1989. The Synth package solves the same nested problem with a different optimizer; the pre-treatment MSPE is flat near its minimum, so the predictor weights V differ across optimizers while the country weights agree to about 0.04. The country weights are what the paper interprets.

```
set.seed(1)
sc_sweden <- synth_control(carbon, id = "country", time = "year", y = "CO2_transport_capita", d = "treated",
  x = xs, lags = c(1989, 1980, 1970), pre_window = 1980:1989, v = "mspe",
  v_control = list(maxit = 2000, starts = 5))

sc_sweden
#> Synthetic control: average post-treatment effect -0.2965
#> 1 treated and 14 donor units; 30 pre- and 16 post-treatment periods; 7 predictors (V: mspe)
#> Pre-treatment RMSPE 0.0347; weight concentration ||w||^2 = 0.246 (4.07 effective donors)
#> Weights above 0.001: Denmark (0.387), New Zealand (0.217), Belgium (0.167), Greece (0.116), United States (0.087), Switzerland (0.087)
```

Table 1 reports the predictor means, and its purpose is to show that synthetic Sweden resembles Sweden on the drivers of emissions while the donor average does not. The paper's last

column is a population-weighted average of the donors; the replication data do not include population, so the column here is unweighted, as in the RTutor version. GDP per capita, vehicles, urban population, and the three emission lags match closely. Gasoline consumption does not, and the paper attributes that to the small weight V assigns to it.

```
bal <- sc_sweden$balance
tab1 <- data.frame(Variables = unname(x_labels[bal$predictor]), Sweden = bal$treated,
  `Synth. Sweden` = bal$synthetic, `OECD sample` = bal$donor_mean, `V matrix` = bal$v, check.names = FALSE)
kable(tab1, digits = c(0, 3, 3, 3, 3), booktabs = TRUE, row.names = FALSE,
  caption = "CO2 emissions from transport, predictor means before the tax reform (paper's Table 1). The OECD column is an unweighted donor average.")
```

Table 1: CO2 emissions from transport, predictor means before the tax reform (paper's Table 1). The OECD column is an unweighted donor average.

Variables	Sweden	Synth. Sweden	OECD sample	V matrix
GDP per capita	20121.479	19531.166	18706.663	0.000
Motor vehicles (per 1,000 people)	405.562	405.563	402.368	0.284
Gasoline consumption per capita	456.178	404.386	425.510	0.000
Urban population	83.099	83.099	75.238	0.714
CO2 from transport per capita 1989	2.506	2.482	2.371	0.001
CO2 from transport per capita 1980	2.005	2.033	2.121	0.001
CO2 from transport per capita 1970	1.729	1.667	1.637	0.000

Table 2 reports the country weights, and the paper reads them as a plausibility check: Denmark, Belgium, and New Zealand, which resemble Sweden in institutions, GDP, and geography, make up three quarters of synthetic Sweden. The paper's weights are Denmark 0.384, Belgium 0.195, New Zealand 0.177, Greece 0.090, the United States 0.088, and Switzerland 0.061, with the rest at or near zero.

```
w <- sc_sweden$weights
published <- c(Australia = 0.001, Belgium = 0.195, Canada = 0, Denmark = 0.384, France = 0, Greece = 0.090,
  Iceland = 0.001, Japan = 0, `New Zealand` = 0.177, Poland = 0.001, Portugal = 0, Spain = 0,
  Switzerland = 0.061, `United States` = 0.088)
tab2 <- data.frame(Country = w$country, Weight = round(w$weight, 3), Published = published[w$country])
kable(cbind(tab2[1:7, ], tab2[8:14, ]), booktabs = TRUE, row.names = FALSE,
  caption = "Country weights in synthetic Sweden (paper's Table 2), replication against published values.") %>% tab_style()
```

Table 2: Country weights in synthetic Sweden (paper's Table 2), replication against published values.

Country	Weight	Published	Country	Weight	Published
Australia	0.000	0.001	Japan	0.000	0.000
Belgium	0.167	0.195	New Zealand	0.217	0.177
Canada	0.000	0.000	Poland	0.000	0.001
Denmark	0.387	0.384	Portugal	0.000	0.000
France	0.000	0.000	Spain	0.000	0.000
Greece	0.116	0.090	Switzerland	0.027	0.061
Iceland	0.000	0.001	United States	0.087	0.088

```
c(max_abs_difference = max(abs(tab2$Weight - tab2$Published)), pre_rmspe = sc_sweden$pre_rmspe,
  concentration = unname(sc_sweden$concentration[["12"]]))
#> max_abs_difference      pre_rmspe      concentration
#>      0.04000000      0.03471442      0.24594823
```

Figure 4 is the identification argument in one picture: synthetic Sweden tracks Sweden for thirty years and departs from it after 1990. The paper reports an average absolute pre-treatment difference of about 0.03 tons. The post-treatment distance is the estimate.

```
paths <- sc_sweden$effects %>% transmute(year = time, Sweden = treated, `Synthetic Sweden` = synthetic) %>%
  pivot_longer(-year, names_to = "series", values_to = "y")
plot_paths(paths)
```

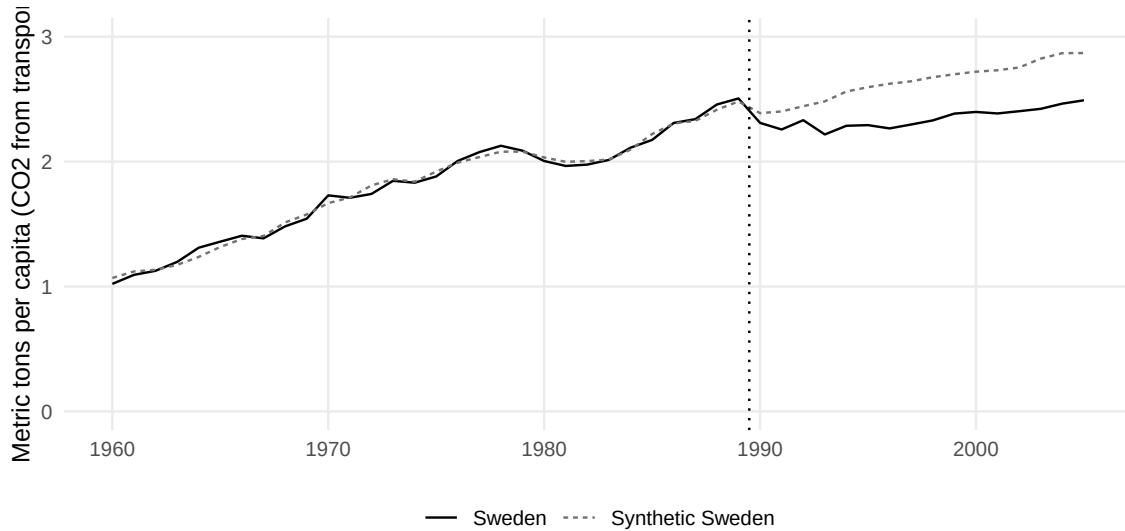


Figure 3: Per-capita CO2 emissions from transport, Sweden versus synthetic Sweden (paper's Figure 4).

Figure 5 plots the gap, and the paper reads three numbers off it: the 2005 reduction, the average reduction, and the reduction in 2000, the last year before the energy tax was lowered. The published values are 12.5 percent (0.35 tons) in 2005, 10.9 percent (0.29 tons) on average over 1990 to 2005, and 11.5 percent (0.31 tons) in 2000. Percentages are gaps relative to the synthetic level.

```
eff <- sc_sweden$effects
ggplot(eff, aes(x = time, y = gap)) + geom_line() + geom_hline(yintercept = 0, linetype = "dotted") + tax_line +
  scale_y_continuous(limits = c(-0.4, 0.4)) +
  labs(x = NULL, y = "Gap in metric tons per capita") + rep_theme()
```

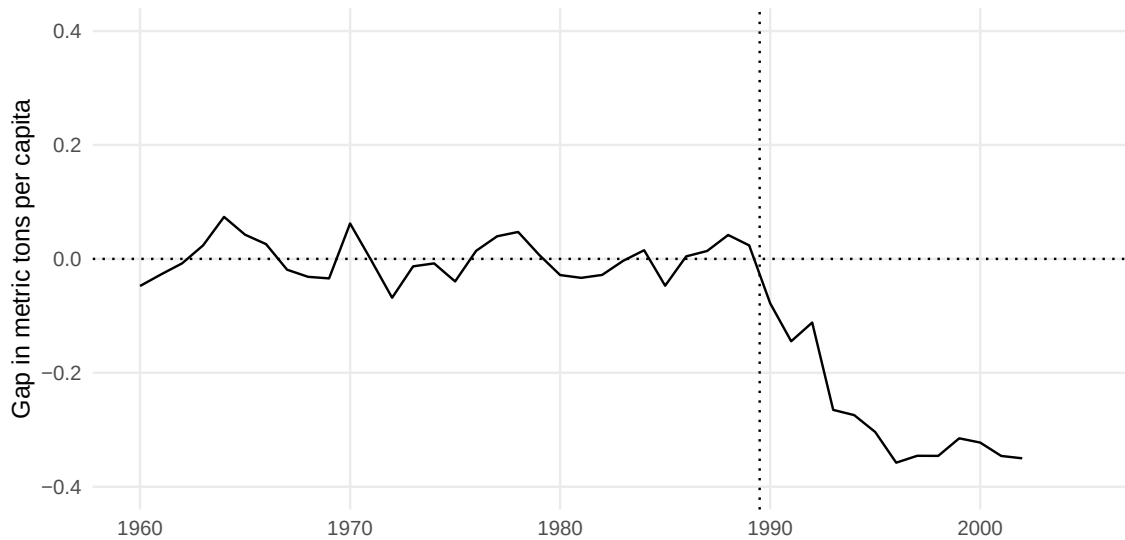


Figure 4: Gap in per-capita CO2 emissions from transport between Sweden and synthetic Sweden (paper's Figure 5).

```
headline <- eff %>% filter(post) %>%
  summarise(average_gap = mean(gap), average_pct = mean(gap / synthetic),
    gap_2005 = gap[time == 2005], pct_2005 = gap[time == 2005] / synthetic[time == 2005],
    gap_2000 = gap[time == 2000], pct_2000 = gap[time == 2000] / synthetic[time == 2000],
    pre_abs_diff = mean(abs(eff$gap[!eff$post])))
round(as.data.frame(headline), 3)
#>   average_gap average_pct gap_2005 pct_2005 gap_2000 pct_2000 pre_abs_diff
#> 1    -0.297    -0.111   -0.378   -0.132   -0.322   -0.119     0.029
```

The replication gives an average reduction of -0.30 tons (11.1 percent), a 2005 reduction of -0.38 tons (13.2 percent), and a 2000 reduction of -0.32 tons (11.9 percent), against the paper's 0.29 (10.9 percent), 0.35 (12.5 percent), and 0.31 (11.5 percent).

5 Placebo tests

The in-time placebos (Figure 6) move the tax to 1980 and to 1970 and ask the method to find an effect where there was none. Each placebo uses only the data before 1990, shorter predictor windows, and emission lags shifted back, following the paper's code: for 1980, covariates over 1970 to 1979 and lags in 1979, 1970, and 1965; for 1970, covariates over 1960 to 1969, lags in 1969, 1965, and 1960, and Poland dropped because its GDP is missing before 1970.

```
set.seed(1)
pt_1980 <- synth_placebo(sc_sweden, type = "time", placebo_time = 1980, lags = c(1979, 1970, 1965), pre_window = 1970:1979)
sc_no_poland <- synth_control(carbon[carbon$country != "Poland", ], id = "country", time = "year",
  y = "CO2_transport_capita", d = "treated", x = xs, lags = c(1989, 1980, 1970),
  pre_window = 1980:1989, v = "mspe", v_control = list(maxit = 2000, starts = 5))
pt_1970 <- synth_placebo(sc_no_poland, type = "time", placebo_time = 1970, lags = c(1969, 1965, 1960), pre_window = 1960:1969)
placebo_paths <- bind_rows(
  pt_1980$effects %>% transmute(panel = "Panel A. 1980", year = time, Sweden = treated, `Synthetic Sweden` = synthetic),
  pt_1970$effects %>% transmute(panel = "Panel B. 1970", year = time, Sweden = treated, `Synthetic Sweden` = synthetic)
) %>% pivot_longer(c(Sweden, `Synthetic Sweden`), names_to = "series", values_to = "y")
ggplot(placebo_paths, aes(x = year, y = y, colour = series, linetype = series)) +
  geom_line() + facet_wrap(~ panel) +
  geom_vline(data = data.frame(panel = c("Panel A. 1980", "Panel B. 1970"), x = c(1979.5, 1969.5)),
    aes(xintercept = x), linetype = "dotted") +
  scale_colour_manual(values = c("black", "grey45")) + scale_y_continuous(limits = c(0, 3)) +
  labs(x = NULL, y = "Metric tons per capita", colour = NULL, linetype = NULL) + rep_theme()
```

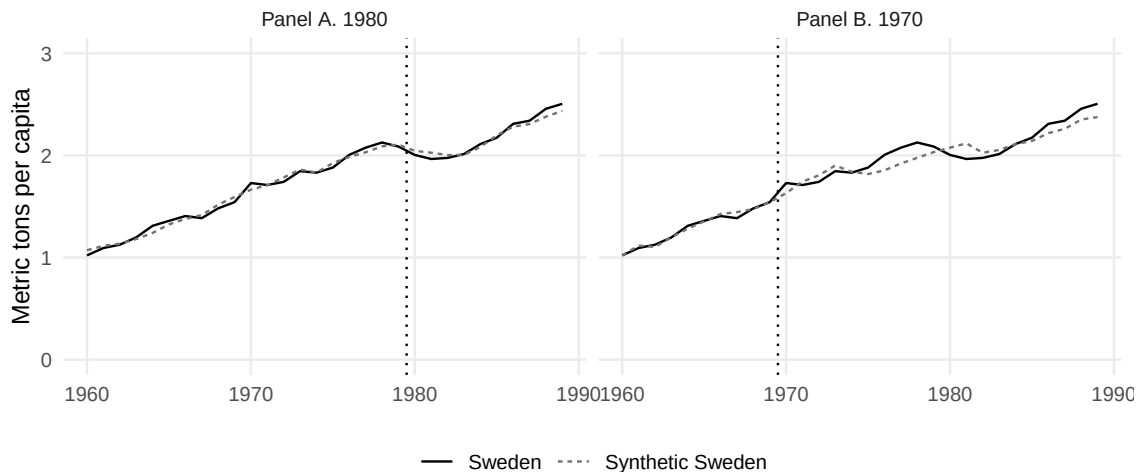


Figure 5: Placebo in-time tests: the tax moved to 1980 (left) and to 1970 (right), estimated on data before 1990 (paper's Figure 6).

```
c(placebo_1980_gap = pt_1980$estimate, placebo_1970_gap = pt_1970$estimate, real_gap = sc_sweden$estimate)
#> placebo_1980_gap placebo_1970_gap      real_gap
```

```
#>      0.00907536      0.03183269     -0.29654392
```

The in-space placebos (Figure 7) reassign the tax to each donor; the paper first shows all 14 gaps and then drops the donors whose pre-treatment MSPE exceeds 20 times Sweden's, which leaves nine. In the restricted set Sweden's post-1990 gap is the largest, so the permutation p-value is 1/10. Each placebo is fitted with the same predictors and with the other donors as its pool.

```
pl <- synth_placebo(sc_sweden, mspe_limit = 20)
gap_long <- function(keep, panel) {
  g <- pl$gaps[, keep, drop = FALSE]
  data.frame(panel = panel, year = rep(pl$time, ncol(g)), gap = as.vector(g),
    id = rep(colnames(g), each = length(pl$time)),
    series = rep(ifelse(colnames(g) == "Sweden", "Sweden", "Control countries"), each = length(pl$time)))
}
d7 <- bind_rows(gap_long(rep(TRUE, ncol(pl$gaps)), "Panel A. All 14 donors"), gap_long(pl$kept, "Panel B. MSPE within 20 times S
ggplot() +
  geom_line(data = filter(d7, series != "Sweden"), aes(x = year, y = gap, group = id), colour = "grey70") +
  geom_line(data = filter(d7, series == "Sweden"), aes(x = year, y = gap), colour = "black", linewidth = 0.8) +
  facet_wrap(~ panel) + geom_hline(yintercept = 0, linetype = "dotted") + tax_line +
  coord_cartesian(ylim = c(-1, 1)) +
  labs(x = NULL, y = "Gap in metric tons per capita") + rep_theme()
```

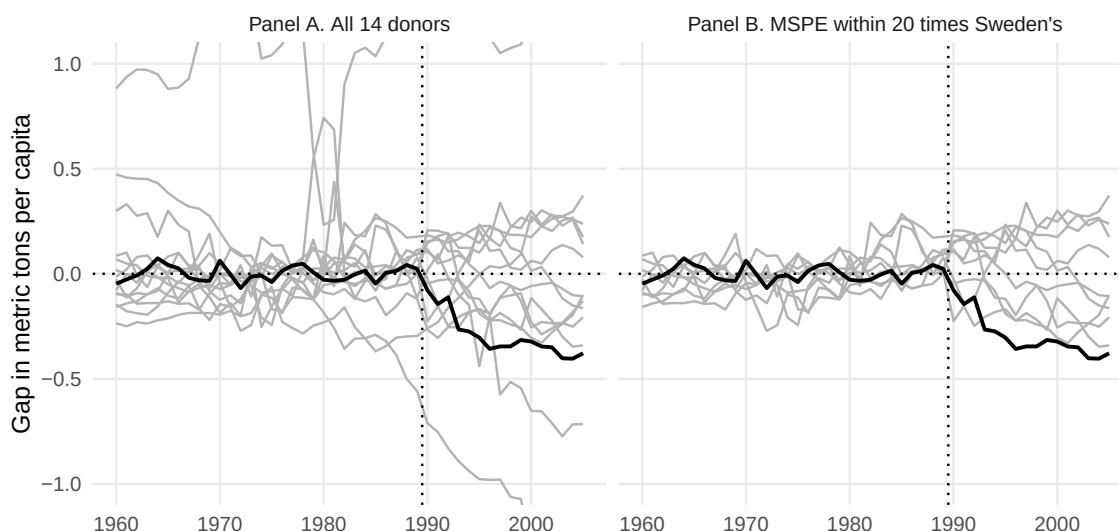


Figure 6: Permutation test: Sweden's gap (black) against the placebo gaps of all 14 donors (left) and of the nine donors with pre-treatment MSPE within 20 times Sweden's (right) (paper's Figure 7).

```
c(kept_after_cut = sum(pl$kept) - 1L, sweden_rank_after_cut = pl$mspe$rank[1L], p_value_after_cut = pl$p_value)
#>      kept_after_cut sweden_rank_after_cut      p_value_after_cut
#>           9.0           1.0           0.1
```

The ratio test (Figure 8) avoids the arbitrary cut by ranking every country on its post-to-pre MSPE ratio; Sweden's ratio is the largest of all 15, so the p-value is 1/15. A large ratio needs both a good pre-treatment fit and a large post-treatment gap, which is why the paper prefers it with few donors.

```
pl_all <- synth_placebo(sc_sweden)
plot_synth(pl_all, type = "ratio")
```

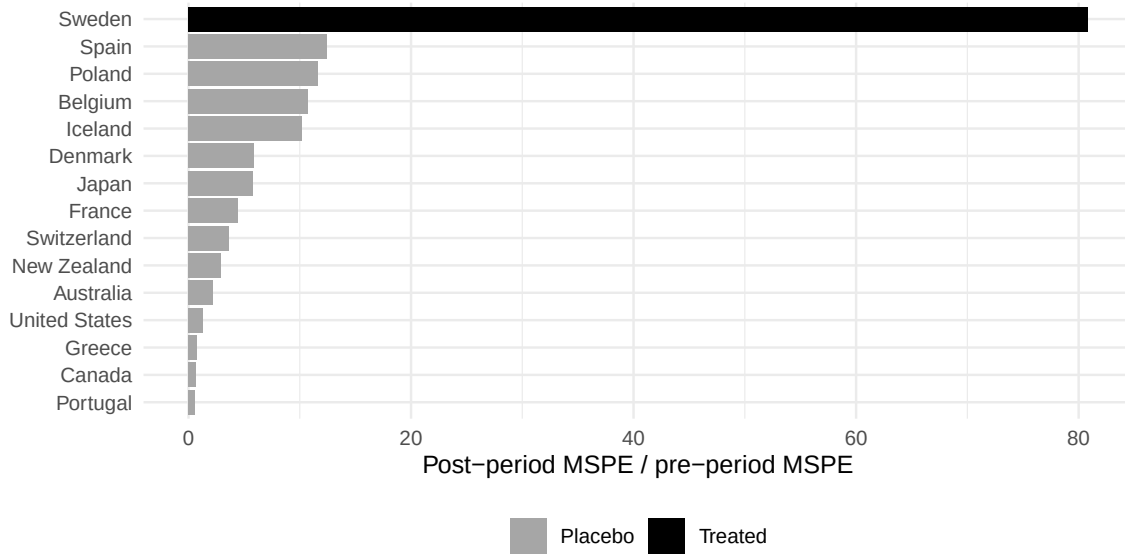


Figure 7: Ratio test: post-treatment MSPE over pre-treatment MSPE for Sweden and the 14 donors (paper's Figure 8).

```
c(sweden_rank = pl_all$mspe$rank[1L], p_value = pl_all$p_value, sweden_ratio = pl_all$mspe$ratio[1L])
#> sweden_rank      p_value sweden_ratio
#> 1.00000000    0.06666667  80.80129167
```

The leave-one-out test (Figure 9) refits synthetic Sweden without each donor that has weight above 0.001; the paper reads the range of reductions, 8.8 percent without Switzerland to 13.0 percent without Denmark, as the sensitivity to any single donor. All refits keep the estimate well away from zero and above the paper's DiD estimate of 8.1 percent.

```
loo <- synth_loo(sc_sweden, min_weight = 0.001)
plot_synth(loo)
```

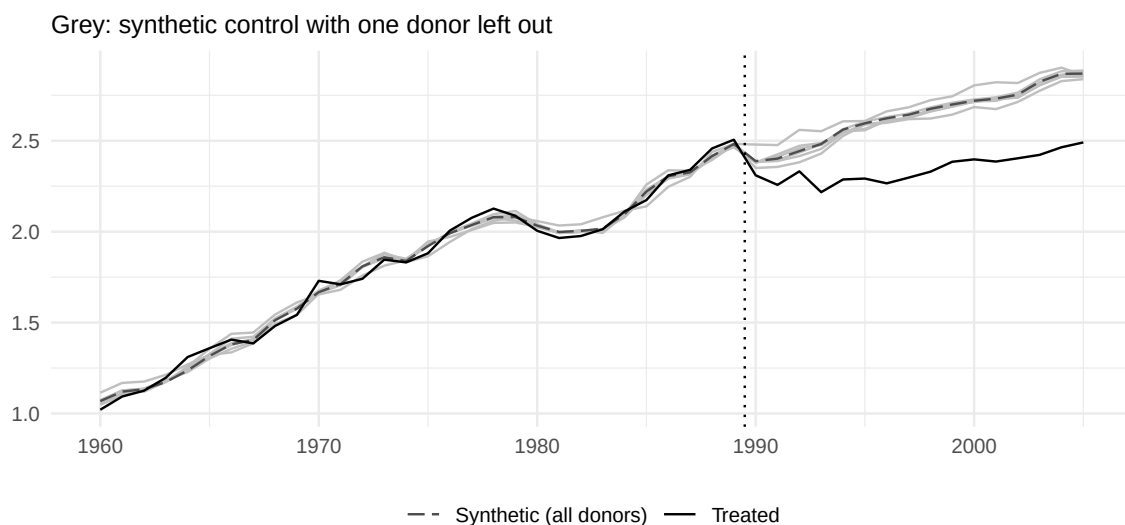


Figure 8: Leave-one-out distribution of synthetic Sweden (paper's Figure 9).

```
loo_tab <- loo$table %>% mutate(pct = sapply(seq_along(loo$fits), function(i) {
  e <- loo$fits[[i]]$effects; mean(e$gap[e$post] / e$synthetic[e$post]) })))
loo_tab %>% mutate(estimate = round(estimate, 3), pct = round(100 * pct, 1), pre_rmspe = round(pre_rmspe, 4)) %>%
```

```
select(dropped, weight, estimate, pct, pre_rmspe)
#>      dropped  weight estimate  pct pre_rmspe
#> 1    Belgium 0.16665131 -0.286 -10.7  0.0366
#> 2    Denmark 0.38659825 -0.352 -13.0  0.0507
#> 3     Greece 0.11625658 -0.286 -10.7  0.0354
#> 4 New Zealand 0.21661170 -0.291 -10.9  0.0360
#> 5  Switzerland 0.02696973 -0.302 -11.2  0.0349
#> 6 United States 0.08691242 -0.256  -9.7  0.0447
```

The full-sample test of the paper (Figure 10) re-estimates with all 24 OECD countries; the replication data contain only the 14 donors, so that figure cannot be reproduced. The paper reports that only the United Kingdom would receive a sizeable weight (0.128) and that the estimate is nearly unchanged.

6 The recession as a possible confounder

Figures 11 and 12 answer the objection that the 1991 to 1993 recession, not the tax, cut emissions. The check reuses the emission weights on GDP per capita. If synthetic Sweden reproduces Sweden's GDP through the recession, the recession is common to both and cannot explain the emissions gap. The paper then compares the two recessions: the GDP gap fell by about 2,300 dollars in both, while the emissions gap rose by 0.086 tons in 1975 to 1978 and fell by 0.291 tons in 1989 to 1993. GDP per capita is missing for Poland before 1970, so the check runs on 1970 to 2005; Poland's weight is zero.

```
gdp <- synth_control(carbon[carbon$year >= 1970, ], id = "country", time = "year", y = "GDP_per_capita", d = "treated",
  weights = sc_sweden)
recessions <- data.frame(start = c(1976, 1991), end = c(1978, 1993))
gdp_paths <- gdp$effects %>% transmute(year = time, Sweden = treated, `Synthetic Sweden` = synthetic) %>%
  pivot_longer(~year, names_to = "series", values_to = "y")
ggplot() +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf), fill = "grey85") +
  geom_line(data = gdp_paths, aes(x = year, y = y, colour = series, linetype = series)) + tax_line +
  scale_colour_manual(values = c("black", "grey45")) +
  scale_y_continuous(limits = c(0, 35000), labels = scales::comma) +
  labs(x = NULL, y = "GDP per capita (PPP, 2005 USD)", colour = NULL, linetype = NULL) + rep_theme()
```

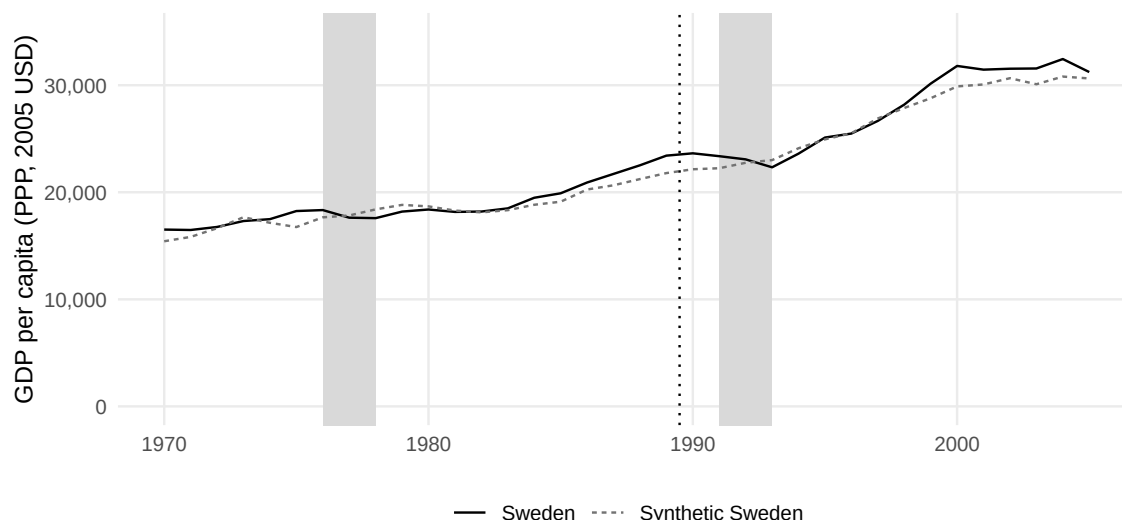


Figure 9: GDP per capita, Sweden versus synthetic Sweden with the emission weights; shaded areas are the 1976 to 1978 and 1991 to 1993 recessions (paper's Figure 11).

```
gaps <- merge(data.frame(year = eff$time, emissions = eff$gap), data.frame(year = gdp$effects$time, gdp = gdp$effects$gap), by =
ggplot() +
```

```
geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf), fill = "grey85") +
geom_hline(yintercept = 0, linetype = "dotted") + tax_line +
geom_line(data = gaps, aes(x = year, y = emissions, colour = "CO2 emissions (left axis)")) +
geom_line(data = gaps, aes(x = year, y = gdp / 5000, colour = "GDP per capita (right axis)")) +
scale_y_continuous(limits = c(-0.4, 0.4), sec.axis = sec_axis(~ . * 5000, name = "Gap in GDP per capita (PPP, 2005 USD)", label = "Gap in GDP per capita (PPP, 2005 USD)")) +
scale_colour_manual(values = c("black", "#B22222")) +
labs(x = NULL, y = "Gap in metric tons per capita", colour = NULL) + rep_theme()
```

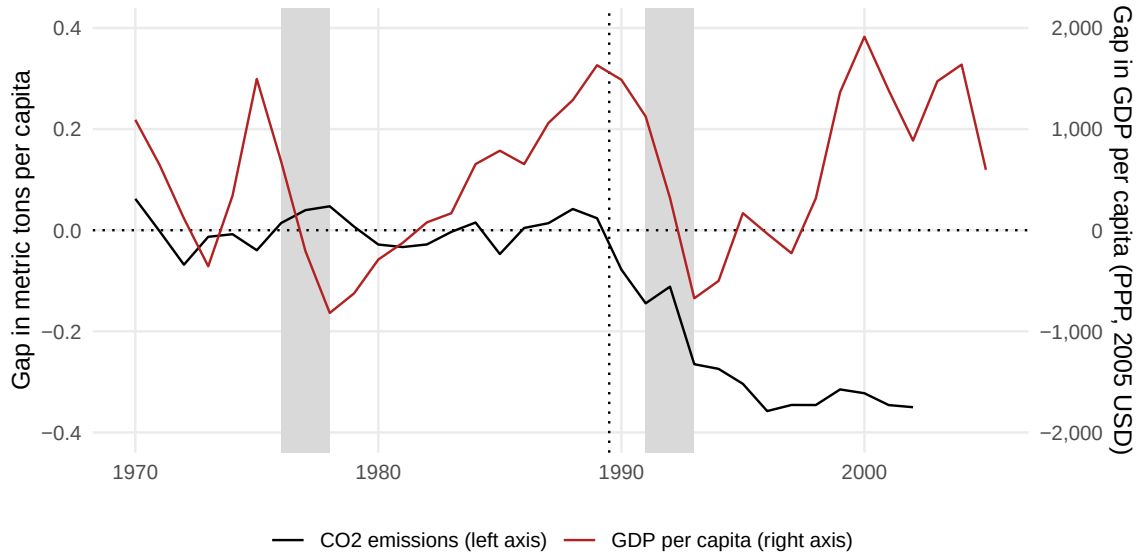


Figure 10: Gaps between Sweden and synthetic Sweden in CO2 emissions (left axis) and GDP per capita (right axis) (paper's Figure 12).

```
change <- function(v, from, to) v[gaps$year == to] - v[gaps$year == from]
data.frame(recession = c("1975 to 1978", "1989 to 1993"),
           gdp_gap_change = c(change(gaps$gdp, 1975, 1978), change(gaps$gdp, 1989, 1993)),
           emissions_gap_change = c(change(gaps$emissions, 1975, 1978), change(gaps$emissions, 1989, 1993)),
           gdp_gap_post_average = mean(gaps$gdp[gaps$year >= 1990]) / mean(gdp$effects$synthetic[gdp$effects$post]))
#>      recession gdp_gap_change emissions_gap_change gdp_gap_post_average
#> 1 1975 to 1978    -2314.223      0.08693153      0.0260986
#> 2 1989 to 1993   -2304.758     -0.28876729      0.0260986
```

7 Disentangling the carbon tax and the VAT

Table 3 estimates the gasoline demand response separately to the carbon-tax-exclusive price and to the carbon tax, on Swedish annual data for 1970 to 2011. The model is log gasoline consumption per capita on the real price excluding the carbon tax (with VAT), the real carbon tax (with VAT), a dummy for 1991 onward, a trend, and, across columns, GDP per capita, urbanization, and unemployment. Columns 5 and 6 instrument the price with the real energy tax and with the crude oil price. Standard errors are Newey-West with 16 lags. The table exists to show that the carbon tax coefficient is about three times the price coefficient, and that the difference survives controls and instruments.

```
reg42 <- reg %>% filter(year >= 1970, year <= 2011, !is.na(log_gas_cons))
f1 <- log_gas_cons ~ p_real_vat + real_CO2_tax_vat + d_CO2_tax + t
f2 <- update(f1, . ~ . + gdp_cap)
f3 <- update(f2, . ~ . + urban_pop)
f4 <- update(f3, . ~ . + unempl)
ols <- list(lm(f1, reg42), lm(f2, reg42), lm(f3, reg42), lm(f4, reg42))
iv_en <- ivreg(log_gas_cons ~ p_real_vat + real_CO2_tax_vat + d_CO2_tax + t + gdp_cap + urban_pop + unempl |
              real_CO2_tax_vat + d_CO2_tax + t + gdp_cap + urban_pop + unempl + real_en_tax_vat, data = reg42)
iv_oil <- ivreg(log_gas_cons ~ p_real_vat + real_CO2_tax_vat + d_CO2_tax + t + gdp_cap + urban_pop + unempl |
              real_CO2_tax_vat + d_CO2_tax + t + gdp_cap + urban_pop + unempl + oil_p_real, data = reg42)
```

```

models <- c(ols, list(iv_en, iv_oil))
nw <- lapply(models, function(m) NeweyWest(m, lag = 16, prewhite = FALSE))
weak_f <- sapply(list(iv_en, iv_oil), function(m) summary(m, diagnostics = TRUE)$diagnostics["Weak instruments", "statistic"])
equal_p <- sapply(seq_along(models), function(i) {
  b <- coef(models[[i]]); V <- nw[[i]]
  d <- b[["p_real_vat"]] - b[["real_CO2_tax_vat"]]
  v <- V["p_real_vat", "p_real_vat"] + V["real_CO2_tax_vat", "real_CO2_tax_vat"] - 2 * V["p_real_vat", "real_CO2_tax_vat"]
  2 * pnorm(-abs(d / sqrt(v)))
})
rows <- data.frame(term = c("p-value: beta1 = beta2", "Instrument F-statistic"),
  rbind(formatC(equal_p, digits = 3, format = "f"), c(rep("", 4), formatC(weak_f, digits = 2, format = "f"))))
names(rows) <- c("term", paste0("(", 1:6, ")"))
modelsummary::modelsummary(
  setNames(models, c("OLS (1)", "OLS (2)", "OLS (3)", "OLS (4)", "IV EnTax (5)", "IV OilPrice (6)")),
  vcov = nw, stars = FALSE, fmt = 4,
  coef_map = c(p_real_vat = "Gas price with VAT", real_CO2_tax_vat = "Carbon tax with VAT", d_CO2_tax = "Dummy carbon tax",
    t = "Trend", gdp_cap = "GDP per capita", urban_pop = "Urban population", unempl = "Unemployment rate",
    `(Intercept)` = "Constant"),
  gof_map = list(list(raw = "nobs", clean = "Observations", fmt = 0), list(raw = "r.squared", clean = "R2", fmt = 2)),
  add_rows = rows, output = "kableExtra",
  title = "Estimation results from gasoline consumption regressions (paper's Table 3). Newey-West standard errors with 16 lags in parentheses",
  tab_style()
)

```

Table 3: Estimation results from gasoline consumption regressions (paper's Table 3). Newey-West standard errors with 16 lags in parentheses.

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	IV EnTax (5)	IV OilPrice (6)
Gas price with VAT	-0.0575 (0.0227)	-0.0598 (0.0190)	-0.0612 (0.0148)	-0.0603 (0.0106)	-0.0620 (0.0199)	-0.0641 (0.0140)
Carbon tax with VAT	-0.2598 (0.0396)	-0.2322 (0.0458)	-0.2335 (0.0480)	-0.1856 (0.0383)	-0.1857 (0.0378)	-0.1857 (0.0377)
Dummy carbon tax	0.1087 (0.0372)	0.0604 (0.0569)	0.0633 (0.0557)	0.0999 (0.0598)	0.0977 (0.0704)	0.0949 (0.0594)
Trend	0.0207 (0.0030)	0.0253 (0.0039)	0.0244 (0.0035)	0.0341 (0.0028)	0.0342 (0.0029)	0.0344 (0.0028)
GDP per capita		-0.0011 (0.0007)	-0.0010 (0.0006)	-0.0037 (0.0010)	-0.0037 (0.0010)	-0.0037 (0.0010)
Urban population			0.0127 (0.0685)	0.0301 (0.0606)	0.0313 (0.0643)	0.0329 (0.0581)
Unemployment rate				-0.0242 (0.0052)	-0.0242 (0.0053)	-0.0242 (0.0053)
Constant	6.2277 (0.1569)	6.4066 (0.1314)	5.3724 (5.6616)	4.4068 (4.9000)	4.3132 (5.1522)	4.1982 (4.6934)
Observations	42	42	42	42	42	42
R2	0.72	0.73	0.73	0.76	0.76	0.76
p-value: beta1 = beta2	0.000	0.001	0.001	0.001	0.004	0.001
Instrument F-statistic					4.76	99.10

The coefficients and Newey-West standard errors match the paper's Table 3 to the third decimal in every column. The instrument F-statistics differ: `ivreg`'s diagnostic is the classical first-stage F, while the paper reports 3.57 and 310.9, presumably with heteroskedasticity and autocorrelation robust standard errors; the conclusion is the same, the energy tax is a weak instrument and the oil price is not.

The elasticities follow by multiplying the semi-elasticities of column 4 by the 1990 real price of 8.48 SEK: the paper reports a price elasticity of -0.51 and a carbon tax elasticity of -1.57. The ratio of about three is the paper's explanation for why a simulation using the price elasticity alone understates the effect of the tax.

```

b4 <- coef(ols[[4]])
c(price_semi = b4[["p_real_vat"]], tax_semi = b4[["real_CO2_tax_vat"]],
  price_elasticity = 8.48 * b4[["p_real_vat"]], tax_elasticity = 8.48 * b4[["real_CO2_tax_vat"]],
  ratio = b4[["real_CO2_tax_vat"]] / b4[["p_real_vat"]])
#>      price_semi      tax_semi price_elasticity tax_elasticity
#>      -0.06033585      -0.18559898      -0.51164803      -1.57387933
#>      ratio
#>      3.07609768

```

Figure 13 uses the column 4 semi-elasticities to simulate emissions under three scenarios and reads the carbon tax's share of the reduction off the distances between the lines. The three series are supplied in the replication data: no carbon tax and no VAT, VAT without the carbon tax, and both. Up to 2000 the two taxes contribute about equally; after 2000 the carbon tax increases dominate. The paper then applies the carbon tax's share of the simulated reduction in each year to the synthetic control gap of that year, and attributes 6.3 percent, or 0.17 tons per capita, of the average post-treatment reduction to the carbon tax, 0.15 tons (5.5 percent) in 2000 and 0.27 tons (9.4 percent, three quarters of the total) in 2005.

```

d13 <- disent %>% filter(year >= 1970, year <= 2005) %>%
  select(year, `Carbon tax and VAT` = CarbonTaxandVAT, `No carbon tax, with VAT` = NoCarbonTaxWithVAT,
    `No carbon tax, no VAT` = NoCarbonTaxNoVAT) %>%
  pivot_longer(-year, names_to = "scenario", values_to = "y")
ggplot(d13, aes(x = year, y = y, colour = scenario, linetype = scenario)) + geom_line() + tax_line +
  scale_colour_manual(values = c("black", "grey45", "#B22222")) +
  labs(x = NULL, y = "Metric tons per capita (CO2 from transport)", colour = NULL, linetype = NULL) + rep_theme()

```

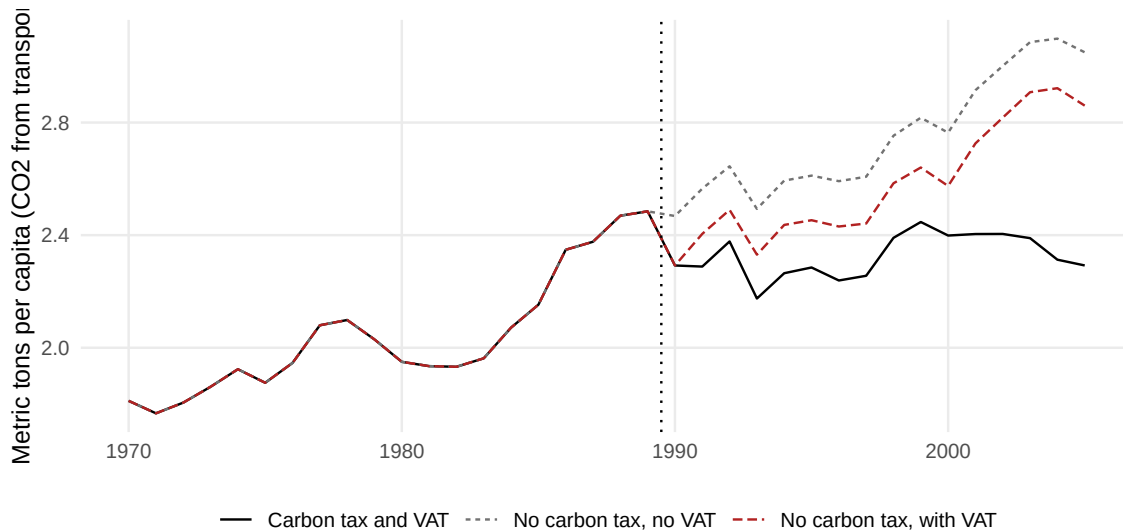


Figure 11: Simulated per-capita CO2 emissions from transport under three tax scenarios, 1970 to 2005 (paper's Figure 13).

```

carbon_share <- disent %>% filter(year >= 1990, year <= 2005) %>%
  transmute(year, share = (CarbonTaxandVAT - NoCarbonTaxWithVAT) / (CarbonTaxandVAT - NoCarbonTaxNoVAT)) %>%
  left_join(efc %>% transmute(year = time, gap, synthetic), by = "year") %>%
  mutate(carbon_tax_gap = share * gap, carbon_tax_pct = carbon_tax_gap / synthetic)
carbon_only <- carbon_share %>%
  summarise(average_gap = mean(carbon_tax_gap), average_pct = mean(carbon_tax_pct),
    gap_2000 = carbon_tax_gap[year == 2000], pct_2000 = carbon_tax_pct[year == 2000],
    gap_2005 = carbon_tax_gap[year == 2005], pct_2005 = carbon_tax_pct[year == 2005], share_2005 = share[year == 2005])
round(as.data.frame(carbon_only), 3)
#>   average_gap average_pct gap_2000 pct_2000 gap_2005 pct_2005 share_2005
#> 1      -0.173      -0.064      -0.155      -0.057      -0.284      -0.099        0.75

```

Figure 14 puts the simulation next to the synthetic control gap; the two agree through 2000

and diverge afterwards, because the simulation holds the energy tax fixed while the real tax reform lowered it as the carbon tax rose. The paper reads the 2005 divergence, -0.757 against -0.355, as what the carbon tax increases would have done had the energy tax not been cut.

```
d14 <- disent %>% transmute(year, `Synthetic control result` = CO2_reductions_synth, `Simulation result` = CO2_reductions_simulation)
pivot_longer(~year, names_to = "series", values_to = "y")
ggplot() +
  geom_rect(data = data.frame(start = 2000, end = 2005), aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf), fill = "grey85") +
  geom_line(data = d14, aes(x = year, y = y, colour = series, linetype = series)) +
  geom_hline(yintercept = 0, linetype = "dotted") + tax_line +
  scale_colour_manual(values = c("black", "#B22222")) +
  labs(x = NULL, y = "Gap in metric tons per capita", colour = NULL, linetype = NULL) + rep_theme()
```

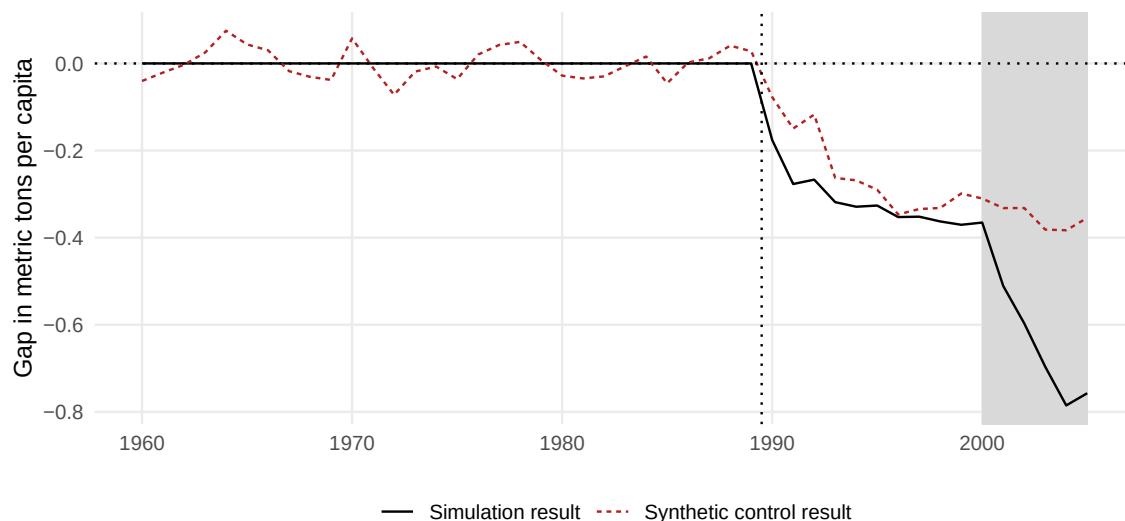


Figure 12: Gap in per-capita CO2 emissions from transport: synthetic control versus simulation (paper's Figure 14).

```
disent %>% filter(year %in% c(2000, 2005)) %>% select(year, CO2_reductions_synth, CO2_reductions_simulation)
#>   year CO2_reductions_synth CO2_reductions_simulation
#> 1 2000         -0.3101433          -0.3653464
#> 2 2005         -0.3546280          -0.7571365
```

8 Checks the paper did not run

Ferman and Pinto (2021) showed that a synthetic control with an imperfect pre-treatment fit is biased when the weights miss the treated unit's level or when assignment is correlated with the factors, and recommended four checks; Sweden passes them. The demeaned synthetic control matches trends only and gives almost the same estimate, so the level is matched. The fit remains close after subtracting the period mean of all countries. The weights are spread over four effective donors. And the specification test does not find the demeaned synthetic control and DiD to disagree.

```
sc_all_lags <- synth_control(carbon, "country", "year", "CO2_transport_capita", "treated")
sc_demeaned <- synth_control(carbon, "country", "year", "CO2_transport_capita", "treated", demean = TRUE)
did <- sdid_weights(carbon, "country", "year", "CO2_transport_capita", "treated", estimator = "did")
spec <- synth_spec_test(sc_demeaned)
conf <- synth_conformal(sc_all_lags, per_period = TRUE)
data.frame(estimator = c("Andersson's specification", "All 30 lags", "Demeaned, all lags", "DiD"),
  estimate = c(sc_sweden$estimate, sc_all_lags$estimate, sc_demeaned$estimate, did$estimate),
  pre_rmspe = c(sc_sweden$pre_rmspe, sc_all_lags$pre_rmspe, sc_demeaned$pre_rmspe, NA),
  concentration = c(sc_sweden$concentration[["12"]], sc_all_lags$concentration[["12"]], sc_demeaned$concentration[["12"]], did$concentration[["12"]]))
#>   estimator estimate pre_rmspe concentration
#> 1 Andersson's specification -0.2965439 0.03471442 0.24594823
#> 2 All 30 lags -0.2836777 0.03430629 0.25185214
```

```
#> 3      Demeaned, all lags -0.2914985 0.03417556 0.26477903
#> 4      DiD -0.2137205      NA      0.07142857
c(demeaned_intercept = sc_demeaned$intercept, spec_test_p = spec$p_value, conformal_joint_p = conf$p_value,
  per_period_rejections_at_5pct = sum(conf$per_period$p_value <= 0.05))
#>      demeaned_intercept      spec_test_p
#>      0.05684003      0.13043478
#>      conformal_joint_p per_period_rejections_at_5pct
#>      0.39130435      14.00000000

plot_synth(sc_sweden, type = "demeaned_path")
```

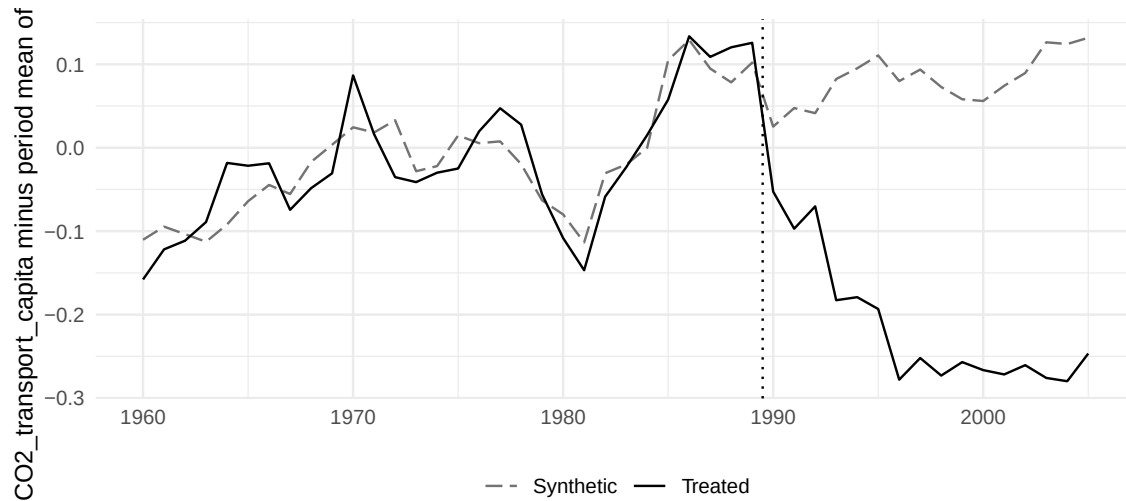


Figure 13: Sweden and synthetic Sweden after subtracting each year’s mean over all 15 countries: the fit the weights achieved beyond the common trend (Ferman and Pinto’s Figure 2.C for this application).

9 Conclusion

The replication reproduces the paper’s estimates and every robustness check within the tolerance one expects from a different optimizer. The country weights agree with Table 2 to within 0.04, the average post-treatment reduction is 11.1 percent against the paper’s 10.9 percent, the in-time placebos find nothing, the ratio test puts Sweden first among 15, the leave-one-out range brackets the estimate, and the recession comparison reproduces the paper’s numbers to three decimals. Table 3 and the disentangling follow from the supplied series.

What the package adds is the imperfect-fit reading. The demeaned estimate, the detrended fit, the weight concentration, and the specification test are the checks a referee would now ask for, and they support the paper’s design.

Session information

```
sessionInfo()
#> R version 4.5.1 (2025-06-13)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.6.2
#>
#> Matrix products: default
#> BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8
```

```

#>
#> time zone: America/Chicago
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods   base
#>
#> other attached packages:
#> [1] ivreg_0.6-8      lmtest_0.9-40    zoo_1.8-14
#> [4] sandwich_3.1-1  haven_2.5.5      causalmetrics_0.0.1
#> [7] testthat_3.3.2  kableExtra_1.4.0 modelsummary_2.4.0
#> [10] broom_1.0.8      estimatr_1.0.6   fixest_0.12.1
#> [13] data.table_1.17.6 lubridate_1.9.4  forcats_1.0.0
#> [16] stringr_1.6.0    dplyr_1.2.0      purrr_1.2.1
#> [19] readr_2.1.5      tidyr_1.3.2      tibble_3.3.1
#> [22] ggplot2_4.0.2    tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#> [1] rlang_1.2.0      magrittr_2.0.4    dreamerr_1.5.0
#> [4] compiler_4.5.1   systemfonts_1.3.1 vctrs_0.7.1
#> [7] quadprog_1.5-8   pkgconfig_2.0.3   fastmap_1.2.0
#> [10] backports_1.5.0  ellipsis_0.3.3    labeling_0.4.3
#> [13] rmarkdown_2.30   sessioninfo_1.2.3 tzdb_0.5.0
#> [16] tinytex_0.58     xfun_0.56         cachem_1.1.0
#> [19] tinytable_0.9.0  stringmagic_1.2.0 parallel_4.5.1
#> [22] R6_2.6.1         tables_0.9.31     stringi_1.8.7
#> [25] RColorBrewer_1.1-3 parallelly_1.45.1 car_3.1-3
#> [28] pkgload_1.5.2    brio_1.1.5        numDeriv_2016.8-1.1
#> [31] Rcpp_1.1.1       bookdown_0.46     knitr_1.51
#> [34] future.apply_1.20.0 usethis_3.2.1     parameters_0.26.0
#> [37] Matrix_1.7-3     timechange_0.3.0  tidyselect_1.2.1
#> [40] rstudioapi_0.17.1 abind_1.4-8       yaml_2.3.12
#> [43] codetools_0.2-20 listenv_0.9.1     pkgbuild_1.4.8
#> [46] lattice_0.22-7   bayestestR_0.16.0 withr_3.0.2
#> [49] S7_0.2.1         evaluate_1.0.5    future_1.67.0
#> [52] desc_1.4.3       xml2_1.3.8        texreg_1.39.5
#> [55] pillar_1.11.1    carData_3.0-5     checkmate_2.3.2
#> [58] insight_1.3.0    generics_0.1.4    rprojroot_2.1.1
#> [61] hms_1.1.3        scales_1.4.0      globals_0.18.0
#> [64] glue_1.8.0       tools_4.5.1       fs_2.1.0
#> [67] grid_4.5.1       datawizard_1.1.0  devtools_2.5.2
#> [70] nlme_3.1-168     performance_0.14.0 Formula_1.2-5
#> [73] cli_3.6.6        textshaping_1.0.1 viridisLite_0.4.3
#> [76] svglite_2.2.1    gtable_0.3.6      digest_0.6.39
#> [79] farver_2.1.2     memoise_2.0.1     htmltools_0.5.9
#> [82] lifecycle_1.0.5  httr_1.4.7        MASS_7.3-65

```